Zhankun Luo

Machine Learning Theory & Optimization

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Research Profile

I develop and analyze algorithms for stochastic optimization and statistical learning, with an emphasis on convergence and finite-sample guarantees. My work spans variance reduction and randomized methods, distributed optimization under stochastic constraints, and the dynamics and statistical guarantees of latent-variable learning. My ongoing research investigates oracle-complexity lower and upper bounds for nonconvex optimization under state-dependent heavy-tailed gradient noise.

Education

Purdue University — **Present**
Ph.D. in Electrical and Computer Engineering — *West Lafayette, IN*
Advisor: Prof. Abolfazl Hashemi; Machine Intelligence & Networked Data Science (MINDS) Group.

Purdue University Northwest — **2021**
M.S. in Electrical and Computer Engineering — *Hammond, IN*

Beijing Institute of Technology — **2019**
B.S. in Telecommunication Engineering — *Beijing, China*

Selected Theoretical Contributions

Variance reduction and randomized algorithms

- Developed a unified high-probability analysis of stochastic variance-reduced estimators using a new Freedman-type inequality, including mirror-descent guarantees in Banach spaces and improved oracle complexity for expectation-constrained optimization. paper [5] **[Preprint; NeurIPS reviews: 5 (Accept), 5 (Accept), 4 (Borderline Accept)]**
- Developed unbiased randomized midpoint estimators that correct discretization bias in nonlinear extragradient updates; used antithetic sampling to cancel first-order variance terms and established guarantees for root finding, constrained variational inequalities, and smooth convex-concave games. paper [3]
- Formulated strong antithetic maps and indices to maximize negative dependence while preserving a target law, connecting optimal antithetic coupling to geometric optimal transport for Monte Carlo integration, approximation, and stochastic optimization. slides [2]

Distributed stochastic minimax optimization with constraints

- Designed a single-loop, primal-only Softmax-weighted switching-gradient method for worst-client optimization under stochastic constraints; established convergence under full and partial client participation with logarithmic high-probability dependence. paper poster code [6] **[UAI 2026]**

Dynamics and statistical guarantees for latent-variable models

- Derived explicit population EM updates across all signal-to-noise regimes; proved that noiseless iterates lie on a cycloid, estimated the super-linear convergence exponent, and improved finite-sample statistical error bounds. paper poster [9] **[ICML 2024]**
- Extended the trajectory analysis to unknown mixing weights and regression parameters, establishing linear-to-quadratic population convergence and arbitrary-initialization finite-sample guarantees. paper code [7] **[IEEE TIT R&R; revised manuscript submitted]**
- Characterized overspecified EM with jointly estimated regression parameters and mixing weights, proving distinct linear and sublinear regimes and deriving iteration, sample, and statistical-accuracy guarantees in population and finite samples. paper code [8] **[TMLR 2026]**
- Proved consistency and asymptotic normality with maximum-likelihood covariance for path-integrated score matching in mixed linear regression with unknown mixing weights under suitable diffusion schedules; derived exact fixed-noise decompositions linking the score-matching loss and gradients to EM operators. paper [4]

Stochastic optimization under state-dependent heavy-tailed noise

- Investigating oracle-complexity lower bounds and algorithmic upper bounds for nonconvex optimization under state-dependent bounds on gradient-noise moments. slides [1]

Publications & Manuscripts

[1] **Z. Luo**, A. Fazla, A. Upadhyay, S. B. Moon, B. Shin, and A. Hashemi. "Stochastic Nonconvex Optimization with Heavy-Tailed Noise: Distance and Suboptimality Dependence." *Manuscript in preparation.*

[2] **Z. Luo**, D. Lee, A. Hashemi, and A. Makur. "Generalized Antithetic Variance Reduction: an Optimal Transport Approach." *Manuscript in preparation.*

[3] **Z. Luo**, M. B. Sahin, A. Upadhyay, B. Sharif, and A. Hashemi. "RAMPAGE: RAndomized Mid-Point for debiAsed Gradient Extrapolation." *Preprint*, 2026. Submitted to *International Conference on Artificial Intelligence and Statistics (AISTATS).* paper

[4] **Z. Luo** and A. Hashemi. "Connecting Score Matching, Maximum Likelihood, and Expectation-Maximization in Mixed Linear Regression." *Preprint*, 2026. Submitted to *Transactions on Machine Learning Research (TMLR).* paper

[5] **Z. Luo**, A. Upadhyay, M. B. Sahin, S. B. Moon, A. Makur, and A. Hashemi. "Unified High-Probability Analysis of Stochastic Variance-Reduced Estimation." *Preprint*, 2026. NeurIPS reviews: 5 (Accept), 5 (Accept), 4 (Borderline Accept). paper

[6] **Z. Luo**, A. Upadhyay, S. B. Moon, and A. Hashemi. "First-Order Softmax Weighted Switching Gradient Method for Distributed Stochastic Minimax Optimization with Stochastic Constraints." *Uncertainty in Artificial Intelligence (UAI)*, 2026. paper poster code

[7] **Z. Luo** and A. Hashemi. "Structural Properties, Cycloid Trajectories and Non-Asymptotic Guarantees of EM Algorithm for Mixed Linear Regression." *Revised manuscript submitted to IEEE Transactions on Information Theory following a revise-and-resubmit decision.* paper code

[8] **Z. Luo** and A. Hashemi. "Characterizing Evolution in Expectation-Maximization Estimates for Overspecified Mixed Linear Regression." *Transactions on Machine Learning Research (TMLR)*, 2026. paper code

[9] **Z. Luo** and A. Hashemi. "Unveiling the Cycloid Trajectory of EM Iterations in Mixed Linear Regression." *International Conference on Machine Learning (ICML)*, 2024. paper code

[10] E. Cai, **Z. Luo**, S. Baireddy, J. Guo, C. Yang, and E. J. Delp. "High-Resolution UAV Image Generation for Sorghum Panicle Detection." *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2022. paper video

[11] N. J. Walla, **Z. Luo**, B. Chen, Y. Krotov, and C. Q. Zhou. "Smart Ladle: AI-Based Tool for Optimizing Caster Temperature." *Iron & Steel Technology Conference (AISTech)*, 2021. paper poster

[12] **Z. Luo**. "Structured Light Vision Systems Using a Robust Laser Stripe Segmentation Method." M.S. thesis, Purdue University, 2021. paper

[13] **Z. Luo**, Y. Zhang, and L. Tan. "Multi-Level Random Sample Consensus Method for Improving Structured Light Vision Systems." *IEEE UEMCON*, 2020. paper

[14] Y. Zhang, **Z. Luo**, J. Hou, L. Tan, and X. Guo. "Computer Vision Techniques for Improving Structured Light Vision Systems." *IEEE EIT*, 2020. paper

Research Experience

Research Assistant, MINDS Group, Purdue University — **Aug. 2025 – Present**
PI: Prof. Abolfazl Hashemi — *West Lafayette, IN*

- Design variance-reduced and randomized methods for stochastic minimax and expectation-constrained optimization, with convergence and oracle-complexity guarantees.
- Develop federated-optimization algorithms for full and partial client participation under communication and resource constraints within the NSF project *Learning through the Air: Cross-Layer UAV Orchestration for Online Federated Optimization*.

Research Assistant, VIPER Laboratory, Purdue University — **Aug. 2021 – May 2022**
PIs: Prof. Edward J. Delp and Prof. Fengqing M. Zhu — *West Lafayette, IN*

- Investigated hierarchy-aware learning for fine-grained food recognition in the Technology Assisted Dietary Assessment (TADA) project and generated labeled UAV imagery with image-to-image GANs for the PhenoSorg plant-phenotyping project.

Research Assistant, CIVS, Purdue University Northwest — **Feb. 2020 – May 2021**
PIs: Prof. Chenn Zhou, Prof. Bin Chen, and Prof. Lizhe Tan — *Hammond, IN*

- Built DNN and LightGBM models for the Smart Ladle steel casting-temperature predictor using process data from SQL databases; developed robust multi-camera, multi-laser algorithms for structured-light metrology.

Teaching Experience

Teaching Assistant, Purdue University ECE — **May 2022 – Aug. 2025**
ECE 69500: Optimization for Deep Learning; ECE 20001/20002: EE Fundamentals — *West Lafayette, IN*

- **ECE 69500:** Supported instruction in stochastic gradient methods, generalization, and distributed learning; held office hours and graded homework and exams. slides video
- **ECE 20001/20002:** Led office hours and help sessions, graded assessments, coordinated TA activities, and compiled an ECE 20001 practice-problem collection. problems video

Selected Presentations

Poster Presenter, Midwest Machine Learning Symposium (MMLS 2026), Purdue University, West Lafayette, IN event June 2026

Professional Service

Journal reviewer: IEEE Transactions on Signal Processing; Stochastic Systems.
Conference reviewer: NeurIPS; ICLR; UAI; AISTATS; AAAI; IEEE ISIT; IEEE ICME; CVIS.
Memberships: IEEE Young Professionals; IEEE Signal Processing Society; IEEE Computer Society.

Selected Research Software

`SoftmaxSGM`	Implementation of the UAI 2026 Softmax-weighted switching-gradient method.
`em_overspecified_mlr`	Numerical experiments for the TMLR 2026 analysis of overspecified EM.
`cycloid_em_mlr`	Numerical experiments and visualizations for the ICML 2024 cycloid analysis.
`smart_ladle`	LightGBM casting-temperature predictor with SQL integration, tested and deployed at Steel Dynamics' Butler Division. SDI Butler
`hierarchy_classification`	PyTorch reimplementation of hierarchy-aware food classification and embedding.
`Kmeans-tsne-visualization`	C++/Qt/OpenGL application for interactive visualization of t-SNE embeddings and k-means iterations.

Leadership, Mentoring & Outreach

- **Graduate Student Mentor, English Training in Engineering (ETIE):** designed machine-learning and computer-vision project frameworks and mentored undergraduate research in sequential prediction and 3D reconstruction.
- **Coordinator and Host, Science Communication Contest:** coordinated participants and judges and organized a science communication program serving more than 3,600 students.
- **Team Leader, Mathematical Modeling Contests:** led model development and implementation for national and international competitions, earning a CUMCM Second Prize and an ICM Honorable Mention.

Selected Awards & Honors

NeurIPS Conference Travel Grant	2025
AIST Hunt-Kelly Outstanding Paper Award, third place	2022
AIST Digitalization Applications Technology Best Paper Award	2021
Innovative Practice Outstanding Student (top 5%)	2019
National Third Prize, Energy Saving and Emission Reduction Competition	2017
Interdisciplinary Contest in Modeling, Honorable Mention	2017
Contemporary Undergraduate Mathematical Contest in Modeling, Second Prize	2016

Technical Background

Theory: stochastic, constrained, minimax, and nonconvex optimization; variance reduction; non-asymptotic statistical analysis; EM and latent-variable models; distributed and federated learning; dynamical systems.

Computation: Python, Mathematica, MATLAB, C/C++, SQL, Bash; PyTorch, TensorFlow, NumPy, SciPy, scikit-learn, pandas, OpenCV; Linux, Git, Docker, CMake.

Selected Coursework

Optimization for Deep Learning; Statistical Machine Learning; Probabilistic Machine Learning; Estimation Theory; Reinforcement Learning; Real Analysis; Linear Algebra; Introduction to Mathematical Statistics; Model-Based Image and Signal Processing.

Academic References

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